higher alkanes are produced. This reaction is used to prepare higher alkanes with even number of carbon atoms.

for example:

$$\underset{\text{methyl bromide}}{CH_3\text{-}Br} + 2Na + Br\text{—}CH_3 \xrightarrow[\text{ether}]{\text{dry}} \underset{\text{ethane}}{CH_3\text{-}CH_3} + 2NaBr$$

### iii) Corey- House Mechanism

An alkyl halide and lithium di alkyl copper are reacted to give higher alkane.

for example:

$$\underset{\text{ethyl bromide}}{CH_3CH_2Br} + (CH_3)_2CuLi \downarrow$$

$$CH_3CH_2CH_3 + CH_3Cu + LiBr$$

### 4) Preparation of Alkanes from Grignard reagents

Halo alkanes reacts with magnesium in the presence of dry ethers to give alkyl magnesium halide which is known as Grignard reagents. Here the alkyl group is directly attached to the magnesium metal make it to behave as carbanion. So, any compound with easily replaceable hydrogen reacts with Grignard reagent to give corresponding alkanes.

for example:

$$\underset{\text{chloromethane}}{CH_3\text{—}Cl} + Mg \xrightarrow{\text{Dry ether}} \underset{\text{methyl magnesium chloride}}{CH_3MgCl}$$

$$CH_3MgCl + H_2O \longrightarrow \underset{\text{methane}}{CH_4} + Mg(OH)Cl$$

### Evaluate Yourself

4) Water destroys Grignard reagents why?

5) Is it possible to prepare methane by Kolbe's Electrolytic method

## 13.2.2 Physical Properties:

**Physical Properties**

**physical State**
Alkanes with
$c_1$ to $c_4$ - gases
$c_5$ to $c_{17}$ - liquids
$c_{18}$ and above solids
At room temperature

**Solubility**
- insoluble in $H_2O$
- Soluble in non polar solvents

**Density**
- Less dense than $H_2O$
- Float on top of $H_2O$

**Boiling point**
- Increases as carbon chain length increases
- Decreases with increases in degree of branching

### 1) Boiling Point and Physical state

The boiling point of continuous chain alkanes increases with increases in length of carbon chain roughly about 30°C for every added carbon atom to the chain. Being non polar, alkanes have weak Vanderwal's force which depends upon molecular surface area and hence increases with increase molecular size. We observe that with same number of carbon atoms, straight chain isomers have



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